

RMT-LLM Laboratory — Experiment Report

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Part I — Detailed Results with Explanations

Experiment: Spectral Signature

Scenario:

Language: / Version:

Key Metrics

Metric	Value
n_layers_analyzed	4
max_lambda_max	0.012519393000410352
min_lambda_min	0.00017214117106267862
mean_spectral_gap	0.005858584940683028

RMT Spectral Analysis

Quantity	Value
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Interpretation

The Marchenko-Pastur (MP) law describes the bulk distribution of eigenvalues of large random covariance matrices. When the largest empirical eigenvalue exceeds the MP upper bound, this signals structured (non-random) information — i.e. the model has 'detected' a fact. Below the N_{crit} threshold the model can still self-correct; above it, the Caputo fractional dynamics dominate and the model enters the 'utility trap' described in the project's main monograph.

Part II — Full Task Launch Logs